DESIGN AND EVALUATION OF AN AUGMENTED REALITY STORYBOOK FOR PRIMARY LEARNERS' PLANT GROWTH UNDERSTANDING & STEWARDSHIP

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Young learners often struggle to understand plant-related science concepts when learning is limited to static textbook illustrations and teacher explanation alone. This challenge is especially important in early primary education, where children need concrete, engaging, and age-appropriate learning experiences. Existing AR-based educational tools show promise, but many systems still suffer from interaction difficulties, unstable content alignment, cognitive overload, and limited consideration of children's physical comfort during handheld use. In the proposed system, the overall solution is a handheld AR storybook that combines printed storybook pages, camera input, mobile sensor data, and child interaction to create an interactive learning experience, while the process layer is organized into four coordinated modules: interaction, visualization, narrative, and simulation. To address this gap, this research proposes a child-centered handheld AR storybook for Cambridge Primary Science Grade 1 learners in Sri Lankan international schools, with a specific focus on a Hybrid Interaction Framework that improves usability, comfort, and learning effectiveness. The framework supports dynamic UI adaptation, physical page anchoring, occlusion handling, fatigue-aware interaction flow, and reduction of split attention during AR learning. The system will be developed using Unity 3D, AR Foundation, ARCore, and ARKit, and evaluated through observation, pre- and post-assessment methods, and system interaction logs. Expected outcomes include improved touch accessibility, stable AR interaction, reduced physical fatigue, lower cognitive burden, and stronger learner engagement. Socially, the research supports more accessible and engaging science learning for young children; commercially, it has potential as an interactive educational product for schools and supervised home learning.

Keywords: Augmented Reality, Interactive Storybook, Hybrid Interaction Framework, Grade 1 Science, Child-Centered Learning, Cognitive Load and Split Attention, Gorilla Arm Effect

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TABLE OF CONTENT

DECLARATION ..... **ii**

ABSTRACT ..... **iii**

ACKNOWLEDGEMENT ..... **iv**

TABLE OF CONTENT ..... **v**

LIST OF TABLES ..... **vii**

LIST OF FIGURES ..... **viii**

LIST OF ABBREVIATIONS ..... **ix**

LIST OF APPENDICES ..... **x**

1. INTRODUCTION **1**

1.1 Background and Context **1**

1.2 Literature Review **2**

1.2.1 Design Patterns for Children's Fine Motor Skills in Mobile Interfaces **3**

1.2.2 Occlusion Handling and Spatial Anchoring in Handheld AR **4**

1.2.3 Ergonomics of Handheld AR: The "Gorilla Arm" Effect and Physical Fatigue **5**

1.2.4 Managing Cognitive Load and the Split Attention Effect in Multimodal Learning **6**

1.3 Research Gap **7**

2. OBJECTIVE **12**

2.1 Main Objective **12**

2.2 Specific Objectives **12**

3. METHODOLOGY **14**

4. HIGH-LEVEL SYSTEM ARCHITECTURE **19**

4.1 Overview of the Proposed Solution and Its Components **19**

5. USER REQUIREMENTS **24**

6. COMMERCIALIZATION PLAN **27**

7. BUDGET AND JUSTIFICATION **32**

8. WORK BREAKDOWN STRUCTURE (WBS) 34

9. GANTT CHART 37

10. REFERENCES LIST 39

11. APPENDICES 42

11.1 Appendix A – Survey Instruments **42**

11.2 Appendix B – Detailed Diagrams **43**

B.1 Overall System Architecture **43**

B.2 Individual Component Diagram **43**

11.3 Appendix C – Preliminary UI Mockups **44**

C.1 Planned UI Design Features **44**

C.2 Purpose of Mockups **44**

11.4 Appendix D – Risk Analysis **44**

11.5 Appendix E – Ethical Documents **45**

E.1 Parent / Guardian Consent Form **46**

E.2 Child Assent Form **46**

E.3 Privacy and Confidentiality **46**

LIST OF TABLES

Table 1.1: Research Gap Comparison 11
Table 2.1: Specific Objectives of the Research 13
Table 3.1: Methodology: Adaptive UI Design for Children's Fine Motor Skills 14
Table 3.2: Methodology: AR Occlusion and Spatial Anchoring 15
Table 3.3: Methodology: Handheld AR Ergonomics and Fatigue 16
Table 3.4: Methodology: Managing Cognitive Load in Multimodal Learning 17
Table 6.1: Primary Customer Persona 27
Table 6.2: Secondary Customer Persona 28
Table 6.3: Indirect Customer Persona 28
Table 7.1: Estimated Budget for the Research Project 32
Table 7.2: Budget Justification for the Research Project 33
Table 8.1: Task Breakdown per Member 34
Table 8.2: Timeline Alignment 35
Table 8.3: Balanced Workload Distribution 36
Table 11.1: Observation sheet 42
Table 11.2: Risk Analysis 45

LIST OF FIGURES

Figure 4.1: System Diagram for Components 20
Figure 4.2: Individual Component and Its Parts 21
Figure 9.1: Gantt Chart for the Hybrid Interaction Framework.....37

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
AR	Augmented Reality
ARCore	Augmented Reality Core
ARKit	Augmented Reality Kit
HCI	Human–Computer Interaction
MR	Mixed Reality
UI	User Interface
VR	Virtual Reality

LIST OF APPENDICES

Appendix A – Survey Instruments

Appendix B – Detailed Diagrams

Appendix C – Preliminary UI Mockups

Appendix D – Risk Analysis

Appendix E – Ethical Documents

1. INTRODUCTION

1.1 Background and Context

In early primary education, learners are introduced to foundational plant concepts such as roots, stems, growth, and basic needs. However, plant structure and function are often conceptually difficult at this age because children commonly form incomplete or inaccurate models of how plants live and grow [1]. When instruction relies mainly on static 2D textbooks and verbal explanations, these biological ideas can remain abstract, limiting meaningful visualization and recall.

This challenge is strongly shaped by the learners' development stage. 6-8 years old, primary students typically operate within Piaget's preoperational stage, where thinking is intuitive and benefits from concrete representations and guided interaction rather than abstract reasoning alone [2]. At the same time, early learners show developing fine motor control, which influences how accurately they can perform precise actions such as tapping small User Interface (UI) targets or maintaining stable hand positions [3]. Short attention spans and sensitivity to cognitive overload further increase the need for simple, tangible and scaffolded learning experiences.

There are several stakeholders who are impacted by the problem: the students (conceptual foundations), the teachers (time and engagement), the parents (home support), and the schools moving towards international curricula where experiential learning is emphasized. It is significant to enhance the understanding of science at the early stages because misconceptions at the beginning of the school experience may persist and impact the progress of learning later on [1].

Current conditions in Sri Lanka also indicate readiness for Technology-Enhanced Learning (TEL). National telecom statistics report 17,736,383 smart devices as of December 2024 [4], while Sri Lanka had 12.34 million internet users in January 2024 [5]. Globally, the Augmented Reality (AR) in education market is projected to expand rapidly [6], reflecting growing investment in spatial learning technologies.

Existing solutions such as AR storybooks, marker-based AR textbooks, and mobile AR learning applications have shown learning and engagement benefits in education [7], and plant-focuses AR has been trialed even with grade one learners [8]. Yet, systematic

reviews mention recurring challenges including usability issues, cognitive load, and interface complexity in AR learning environments [9]. For Grade one specifically, many interfaces are not optimized for fine motor limitations, and extended mid-air interaction can cause fatigue [10]. In addition, AR experiences that require learners to alternate between physical materials and a screen can trigger split-attention effects, increasing mental integration demands [11].

In response to these limitations, the proposes research introduces a hybrid AR learning system for Cambridge Grade one plant lessons, integrating physical storytelling with AR overlays, interactive visualization and assessment. Within this broader system, this research focuses on a child-centered hybrid interaction framework that reduces motor burden and cognitive load through dynamic UI resizing, predictive interface adaption, frictionless flow and improved hybrid integration.

This research aligns primarily with Sustainable Development Goal (SDG) 4 (Quality Education) by improving developmentally appropriate early science learning, and contributes indirectly to SDG 13 (Climate Action) and SDG 15 (Life on Land) by building early environmental literacy. It also aligns with Mixed Reality (MR) research cluster through its focus on child-centered AR interaction design and hybrid learning experiences.

1.2 Literature Review

Augment Reality interaction research that combines a printed physical book with device-based AR overlays to support young learners' engagement and conceptual understanding [12]. In education, meta- analyses and literature reviews generally report that AR can improve motivation and learning outcomes, but the effects vary by design quality, learner characteristics, and how well the AR presentation follows learning principles [13]. Importantly, studies that compare AR storybooks with traditional print versions demonstrate that AR can increase interest, but learning benefits depend on whether the AR elements truly support comprehension rather than distract attention [12].

For Cambridge Primary Science Grade 1 learners, this matters because they have limited sustained attention and developing motor control. So, the AR system must be

designed to minimize unnecessary interaction steps and reduce cognitive overload. Research on AR and cognitive load indicates that AR may reduce cognitive overload when information is spatially integrated with task steps, but it can also increase the load if learners must constantly shift gaze, interpret complex 3D scenes or manage competing visual resources [14], [15]. Therefore, the present review focuses on four areas that directly inform the design of hybrid interaction framework: **interaction design, occlusion realism and anchoring stability, physical ergonomics during handheld use and cognitive load management**. Each subsection synthesizes existing contributions, identifies limitations, and highlights the research gap in this study.

1.2.1 Design Patterns for Children's Fine Motor Skills in Mobile Interfaces

Research on children's touchscreen interaction shows consistent performance differences compared to adults. Massive work characterizing children's touches across multiple studies reports that children's input is typically less precise and more variable, leading to higher error rates when targets are small, closely spaced or require continuous control [16]. These findings imply that adult-oriented mobile UI standards can be inappropriate for young learners and that design patterns must explicitly accommodate motor variability through larger touch targets, forgiving hit boxes and simplified gesture demands [17], [16]. Complementing this, research specifically focusing on children in the 3-6 age range demonstrates that even basic tap, drag and drop actions can be challenging and performance degrades when tasks involve multi step gestures or fine control, reinforcing the need for age-appropriate interaction complexity [17].

Recent syntheses highlight that children's touch performance is qualitatively different and that interaction success depends on how applications structure actions. A meta-analysis focuses on pre-operational children's touch gestures summarizes that younger children can reliably perform a limited set of gestures, while more complex multisensory and precision gestures show greater variability and require scaffolding [18]. Empirical work examining how young children learn and operate commercial tablet applications suggests that children can acquire operational skills, but success depends on the apps' interaction demands and the presence of supportive cues [19]. These studies caution that many commercial apps are not designed around development constraints and can mask learning difficulties behind engagement [19]. Studies

evaluating preschool learning apps that combine touch with alternative inputs such as voice suggest that multimodal input can reduce motor burden for some tasks, but only if the design avoids increasing cognitive complexity or adding unreliable recognition steps [19].

Existing research approaches provide valuable measurement of errors, task time and gesture feasibility, but many studies are conducted in controlled tasks that do not reflect authentic classroom contexts and sustained use during learning sessions [17], [18]. In practice, products and services aimed at children's learning often prioritize engaging visuals over motor accessible interaction patterns, resulting in frequent mis-taps, repeated correction cycles and frustration [17], [16]. These limitations motivate the component task of developing dynamic and predictive UI adjustments tailored to grade 1 learners' fine-motor constraints.

1.2.2 Occlusion Handling and Spatial Anchoring in Handheld AR

In AR, occlusion handling and stable anchoring are central to perceived realism and usability. Occlusion methods are commonly grouped into depth-based and model-based approaches [20], [21], [22]. Reviews on model-based occlusion emphasize that while these techniques can improve realism, they often introduce latency, require complex scene understanding or struggle under dynamic conditions [20]. Depth-based approaches can be effective, but they are constrained by device capabilities and depth quality [21], [22]. Empirical work on real-time occlusion in AR guidance contexts proposes monocular reconstruction and depth-map densification to support occlusion without dedicated depth sensors, yet such approaches still face computational cost and robustness limits in real-world scenes [20].

Modern AR platforms expose practical occlusion tools which reduce engineering effort and they can produce convincing results when device support is available [21], [22]. However, platform documentation implicitly reveals important constraints such as depth-based occlusion requires compatible hardware or software and it can degrade in poor feature environments or when depth estimation is uncertain [21], [22]. In research frameworks such as Z-HandAR integrate external depth sensing to improve occlusion quality, but such setups are unrealistic for schools due to cost, setup complexity and portability constraints.

Spatial anchoring research similarly highlights trade-offs among marker-based stability, markerless Simultaneous Localization and Mapping (SLAM)-based placement and persistent anchors [23], [24]. Benchmarking work comparing virtual object stability in mobile AR reports measurable drift and relocalization issues, implying that stable alignment cannot be assumed across devices, sessions or classroom light changes [23]. Broader benchmarking and mapping research for AR further shows that robust localization requires strong visual features and consistent environments [24]. When these conditions are not met, anchors can shift. It is very problematic when overlays must align with a printed page during learning. As a result, the main limitation of existing systems is not that occlusion and anchoring are unsolved, but that their reliability varies sharply with environmental conditions and device constraints [23], [24]. So, this forms a key gap addressed in this study.

1.2.3 Ergonomics of Handheld AR: The "Gorilla Arm" Effect and Physical Fatigue

Ergonomics is a practical barrier to sustained handheld AR use, specially, for children who must hold a device steady while viewing overlays. Studies examining physical demands in AR report that neck and shoulder load, discomfort and fatigue can increase during immersive interaction compared to traditional tasks and that outcomes depend on interaction distance, target size and posture [25]. Recent work discussing real-time AR use highlights fatigue related issues as a known limitation of prolonged raised-arm interaction and device mediated pointing which can reduce session length and willingness to continue [26].

Research on fatigue modeling and prediction shows that fatigue can be anticipated and measured through biomechanical or simulation-based methods, supporting the idea that fatigue risk is designable rather than unavoidable [27]. Meanwhile, prior experimental work demonstrates that redesigning interaction to reduce unsupported gestures can significantly lower perceived exertion, implying that interaction technique is a direct lever for fatigue reduction [27].

Many ergonomic studies use adult participants and lab tasks, which may underrepresent fatigue dynamics for Grade one children who have different strength, operation habits and classroom constraints [25]. Commercial AR leaning products often assume short,

novelty-driven sessions and do not explicitly manage posture, rest cycles or interaction pacing. This becomes a limitation when AR is integrated into somewhat longer science lessons. Therefore, the proposed proposal focuses on designing brief and impactful interaction spurts and reducing sustained arm load directly addresses a gap in both research translation and product design [26], [25].

1.2.4 Managing Cognitive Load and the Split Attention Effect in Multimodal Learning

Cognitive load research in AR education consistently reports that AR can help learning when it integrates information near the relevant real-world referent, but it can also overload learners when it introduces extra visual complexity or frequent attention switching [28], [15]. A meta-analysis focused on AR and cognitive load reflects this ongoing debate, indicating that outcomes vary by design features, learning task type and learner characteristics [15]. Experimental evidence shows that AR presentations can create a spatial split attention effect when key information sources are separated and that can reduce performance and increase cognitive effort [28]. Outside AR specifically, objective, measurements using eye tracking confirm that split attention layouts increase fixation demands and reduce retention compared with integrated designs, supporting the relevance of split attention as a measurable usability and learning risk [29].

Recent AR learning research proposes that the design should follow spatial and temporal contiguity principles to reduce extraneous processing. Controlled studies in instructional contexts show that applying these integration principles in AR can reduce unnecessary cognitive processing and support learning, but only when the AR overlays are restrained and clearly mapped to the task [28]. Broader reviews of AR in science learning also suggest positive effects on achievement and motivation, yet repeatedly note that poorly designed AR increases distraction and can shift effort away from conceptual understanding toward managing the interface [30], [31]. The risk is particularly relevant for primary students, because their working memory capacity and reading fluency are still developing. So, visual clutter, long instructions or multi-step interactions can quickly exceed what they can comfortably manage [29], [30].

Commercial products such as AR picture book experiences and classroom AR tools demonstrate strong engagement, but they frequently emphasize fascinating effects that can unintentionally raise extraneous load if they are not carefully aligned with learning goals [12], [31]. In addition, platform capabilities can increase realism while increasing visual complexity, making it essential to decide what to show, when to show it and how to guide attention [21], [22]. The key limitation across existing systems is that many AR learning designs measure achievement and motivation without providing insufficient analysis of why cognitive load rises or falls and they rarely connect findings to concrete interaction rules [15], [30]. This gap supports the task of evaluating cognitive load, engagement and split attention reduction as design outcomes.

1.3 Research Gap

Overall research problem:

How can primary learners studying the Cambridge Grade 1 Science module in Sri Lanka be better supported in understanding plant growth and environmental changes that are difficult to learn through traditional text-based teaching methods?

Specific research problem:

How do control difficulty, physical fatigue, divided attention, and visual obstruction in handheld AR storybook interactions influence children's usability experience and learning engagement?

Augmented Reality systems have been increasingly used to support early learners by combining printed pages with interactive overlays. Prior educational research indicates that AR can improve learner motivation and achievement, particularly when digital content is aligned with instructional goals and learning principles rather than serving as a distraction [7], [9], [13]. However, evidence shows that the learning impact of AR storybook is inconsistent. AR can increase children's interest, but comprehension gains depend on whether the AR elements are designed to guide attention and support analysis instead of creating additional visual and interaction demands [12]. This is important for Cambridge Primary Science Grade 1 learners in Sri Lanka's international school context, because children at this stage have developing motor control and limited

attention resources. Therefore, the effectiveness of the overall AR storybook system depends heavily on interaction design decisions.

Identified Research Gap 1 – Absence of adaptive UI for Grade 1 motor skills: Existing child-touch research recommends big targets and simple gestures, but most applications still use fixed buttons and do not adapt in real time to 6-year olds' touch variability, causing mis-taps and frustration. [3], [16]- [19]

Although existing studies contribute valuable foundations, the literature reveals four interrelated gaps that directly affect the feasibility and learning value of the proposed hybrid interaction framework. The first gap is the absence of child-centered adaptive mobile interface patters that are explicitly engineered for early learners' fine motor variability during sustained learning tasks. Research consistently shows that young children's touch input is less precise and more variable than adults increasing the error rates when targets are small, closely spaced or require continuous control [16], [17]. While prior work proposes general guidelines such as larger touch targets and simplified gestures, most current approaches evaluate performance in controlled tasks and they do not provide a robust, real-time mechanism that adjusts interaction demands dynamically to children's input behavior [16], [18]. Consequently, many children's learning applications still rely on fixed layouts [19]. Therefore, rather than treating accessibility as static UI sizing, an evidence-driven adaptive interaction approach that operationalizes these findings into a quantifiable design solution for Grade 1 learners is lacking.

Identified Research Gap 2 – Unreliable page-aligned AR in classrooms: Current occlusion and anchoring methods can fail under real classroom conditions such as hands over the book, lighting changes, weak visual features. So, stable alignment with printed pages cannot be guaranteed. [20]- [24]

The second gap concerns the limited robustness of occlusion handling and spatial anchoring in realistic classroom conditions. Existing research demonstrates progress in occlusion techniques such as monocular reconstruction and depth map enhancement, yet these approaches often remain computationally demanding and sensitive to real-world noise and dynamic occluders such as hands [20]. At the same time, commercially

available platform services provide occlusion support via depth estimation, but their performance is constrained by device capabilities and environmental conditions [21], [22]. Anchoring stability research further demonstrates that markerless mobile AR can suffer from drift and relocalization issues across sessions and stable alignment cannot be assumed in practice [23]. Benchmarking work also indicates that strong visual features and consistent environments are required for reliable localization, which is difficult to guarantee in everyday classrooms [24]. Therefore, the key limitation is that existing solutions do not provide the reliability needed for page-aligned learning interactions. This motivates the need for a hybrid strategy that comprehensively addresses hand-book interaction, tracking recovery and anchor stability as design objectives.

Identified Research Gap 3 – Lack of fatigue-aware handheld AR design: Many AR learning tools ignore fatigue and sustained holding discomfort. They rarely include pacing, short interaction cycles or fatigue reduction techniques which can limit session length and usability for children. [10], [25], [26]

The third gap is the lack of fatigue-aware interaction frameworks tailored to handheld AR learning. Ergonomic evidence shows that immersive interaction can increase physical load and discomfort [10], [25]. Even though these results are often reported in adult samples, they are still important because children have lower strength and different handling habits, making fatigue constraints potentially more severe in real classrooms. Research also suggests that fatigue is measurable and can be quantified using interaction metrics [10], [26]. Despite this, many AR educational products implicitly assume short and novelty-driven sessions and do not include pacing, rest-aware interaction cycles or interaction designs that reduce sustained arm elevation. As a result, current approaches remain insufficient for lessons that require repeated engagement, where discomfort can undermine both usability and learning continuity. This creates a clear need for an interaction flow that explicitly minimizes continuous holding, reduces unnecessary motion and structures learning into brief, high impact interaction spurts that are ergonomic by design.

Identified Research Gap 4 – Weak control of cognitive load and split attention: Many AR learning systems increase visual complexity and attention switching and evaluate only engagement without clear design rules to reduce extraneous load and split attention and that is critical for Grade 1 learners. [11], [13]- [15], [29]- [31]

The fourth gap relates to insufficient control of cognitive load and split attention in multimodal AR learning. Evidence from multimedia learning shows that split attention designs increase mental effort when learners must integrate separated information sources and this principle remains highly relevant when AR overlays compete with printed content and on-screen instructions [11]. Research further indicates that cognitive load outcomes are inconsistent [13], [15]. Empirical findings also show that AR presentation choices can alter workload and learning effectiveness, demonstrating that interface design is directly linked to cognitive outcomes [14], [28]. However, many current AR learning systems still evaluate success mainly through achievement or motivation measures without sufficiently explaining why cognitive load fluctuates or translating findings into concrete design rules [30], [31]. This gap is particularly critical for Grade 1 learners whose working memory capacity and reading fluency are still developing. Accordingly, the proposed work addresses this gap by treating cognitive load reduction and split attention mitigation as explicit design and evaluation targets alongside engagement.

Taken together, these gaps justify the novelty and creativity of the proposed hybrid interaction framework which integrates evidence from children's motor development [3], touchscreen interaction research [16]- [18], AR system robustness studies [20], [23], [24], ergonomic and fatigue metrics [10], [25], [26] and cognitive theory and measurement [11], [15], [29]. The evidence indicates that current approaches are insufficient because they either optimize a single aspect in isolation or assume platform defaults will generalize across young learners and classroom variability.

Table 1.1: Research Gap Comparison

Gap	What Existing Studies Address	What is Still Missing	Proposed Contribution
Adaptive child-centered interaction	Studies identify children's fine motor variability and recommend large targets/simple gestures [16], [17], [18]	No real-time adaptive interface mechanism for sustained Grade 1 learning tasks	A quantifiable adaptive interaction model that adjusts to children's touch behavior
Occlusion and anchoring robustness	AR occlusion, depth APIs, and tracking stability have been studied [20], [21], [22], [23], [24]	Reliability remains weak in realistic classroom settings with hands, books, drift, and relocalization issues	A hybrid AR interaction strategy focused on page alignment, recovery, and stability
Fatigue-aware AR learning	Fatigue and workload can be measured in AR interaction [10], [25], [26], [27]	Few educational AR systems reduce sustained holding or structure interaction around ergonomic constraints	A fatigue-aware interaction flow using short, low-strain learning spurts
Cognitive load and split attention control	Multimedia and AR studies show split attention and cognitive load affect learning [11], [15], [28], [29], [30], [31]	Limited translation of findings into concrete design rules for young children using AR storybooks	A design framework that explicitly minimizes split attention and cognitive overload

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to design, develop and evaluate a hybrid interaction framework for handheld AR storybook that improves the usability, comfort and learning effectiveness of Cambridge Primary Science Grade 1 learners in Sri Lankan international schools within the proposed study period.

This objective is based on previous research that demonstrates that AR can improve learning and engagement when the system is designed to support students understand instead of distracting them [7], [9], [12], [13]. Since early primary learners are still developing fine motor control, attention regulation and basic conceptual understanding, the interaction design of the AR system must be carefully aligned with their developmental needs [2], [3], [16]- [19].

Existing studies show that young children experience difficulty with precise touchscreen interaction and complex gesture-based input, indicating the need for simpler and more supportive interaction methods [3], [16]- [18]. At the same time, AR research demonstrates that the educational value of AR systems depends on the stability of virtual content, the clarity of information presentation and the ability of the interface to reduce unnecessary cognitive burden during learning [13]- [15], [20]- [24], [28]- [30]. Prior ergonomic studies also show that prolonged handheld interaction may reduce comfort and usability due to physical fatigue, particularly in immersive or sustained usage settings [10], [25]- [27]. Therefore, the main objective will be achieved by creating and evaluating a framework that responds to these established limitations in current systems and supports more effective, comfortable and developmentally appropriate AR based learning for Grade 1 students.

2.2 Specific Objectives

To achieve the main objective, the study will pursue the following specific objectives. These objectives are designed to break down the research into distinct and measurable components, ensuring systematic development and evaluation of the proposed hybrid interaction framework. Each objective addresses a critical dimension of the system: interaction design, occlusion realism and anchoring stability, physical ergonomics

during handheld use and cognitive load management. By accomplishing these objectives, the project will not only demonstrate the feasibility and effectiveness of the proposed hybrid handheld AR interaction framework, but also provide empirical evidence on how child-centered interaction design, stable AR alignment, ergonomic handheld use and cognitive load reduction can improve the usability and learning experience of Grade 1 science learners.

Table 2.1: Specific Objectives of the Research

No.	Objective
1	**Child-friendly Interaction Design:** Design and implement a child-friendly interaction design for handheld AR storybook use by improving touch accessibility through larger interactive elements, simplified gesture patterns and adaptive interface responses suitable for Grade 1 learners.
2	**Occlusion Handling and Stable Anchoring:** Develop an occlusion handling and spatial anchoring mechanism that improves the visual realism, alignment accuracy, and stability of AR overlays during interaction with printed storybook pages.
3	**Ergonomic Handheld interaction:** Design an ergonomically appropriate handheld AR interaction flow that reduces prolonged arm elevation, unnecessary movement and physical fatigue during learning activities.
4	**Cognitive Load and Split-Attention Reduction:** Apply cognitive load reduction and split attention control principles to the AR learning interface in order to improve clarity, reduce extraneous mental effort and support better learner focus.

3. METHODOLOGY

This study adopts a Design Science Research approach supported by experimental prototype evaluation. This approach is appropriate because the research is not limited to observing an existing problem; instead, it aims to design, develop, and validate a practical interaction framework for a real educational setting. The overall prototype will be implemented as part of the handheld AR storybook system that uses printed pages, camera input, mobile sensors, and touch-based user interaction. Since the study involves child users, the methodology also includes observation-based data collection, interaction logging, and ethical safeguards such as parental consent, child assent, anonymization, and institutional ethical clearance. The methodology is structured around four specific objectives. The detailed methodology for each objective is summarized in Tables 3.1 to 3.4.

Table 3.1: Methodology: Adaptive UI Design for Children's Fine Motor Skills

Aspect	Methodology
Research approach	This subsection uses a design-and-evaluate approach within Design Science Research. First, a child-friendly mobile interaction model will be designed for Grade 1 learners. Then, its effectiveness is evaluated through prototype-based testing.
Technical methods and algorithms	A dynamic user interface mechanism is developed to improve touch accessibility for young learners. The system uses larger touch targets, simplified tap-based actions, and adaptive resizing of buttons based on interaction difficulty or device shake intensity. A lightweight predictive interaction logic is also used to reduce mis-taps and improve selection accuracy.
Tools and platforms with justification	Unity 3D with AR Foundation is used as the main development environment because it supports handheld AR development across Android and iOS devices. **C#** is used for implementing adaptive UI logic and touch handling. Android tablets or smartphones are used for testing because they represent realistic classroom devices.
Data collection and ethical considerations	Data is collected through child interaction sessions, observation notes, and system logs such as touch errors,

	successful selections, repeated attempts, and task completion time. Because the study involves young children, parental consent, child assent, privacy protection, and non-identifiable storage of interaction data is maintained throughout the study.
Validation strategy and performance metrics	The adaptive UI design is validated by comparing interaction performance before and after applying the child-friendly design features. Performance is measured using touch accuracy, number of mis-taps, task completion time, number of retries, and observed ease of interaction. Teacher or facilitator observations may also be used to assess whether the interface is developmentally appropriate.

Table 3.2: Methodology: AR Occlusion and Spatial Anchoring

Aspect	Methodology
Research approach	This subsection follows a prototype-driven technical design and evaluation approach. The aim is to develop and test AR overlay stability and realism when children interact with printed storybook pages.
Technical methods and algorithms	A page-based physical anchoring mechanism will be developed using image targets or marker-based recognition for reliable alignment with storybook pages. In addition, an occlusion handling method is implemented so that hand interactions and tactile contact with the book can occur without incorrectly hiding important virtual content. The system uses page tracking, pose estimation, and layer control logic to maintain stable placement of AR content.
Tools and platforms with justification	Unity 3D, AR Foundation, ARCore, and ARKit is used because they provide support for image tracking, spatial anchoring, and device-level AR rendering. These platforms are suitable for building page-aligned handheld AR applications and are already consistent with the proposed system architecture and team skill base.
Data collection and ethical considerations	Data is collected through system logs and controlled interaction trials, focusing on tracking stability, anchor drift, relocalization behavior, and occlusion errors during physical

	page interaction. Since the users are children, testing sessions ate supervised, short in duration, and designed to avoid frustration or confusion. All recorded data is anonymized.
Validation strategy and performance metrics	Validation focuses on alignment accuracy, tracking stability, recovery time after movement, frequency of anchor drift, and visual correctness of occlusion. Observational feedback is also used to identify whether the virtual content remains understandable and properly aligned during natural interaction with the book.

Table 3.3: Methodology: Handheld AR Ergonomics and Fatigue

Aspect	Methodology
Research approach	This subsection uses an interaction design and usability evaluation approach. The aim is to create a handheld AR interaction flow that reduces physical strain during learning activities.
Technical methods and algorithms	The interaction flow is designed around short, high-impact interaction cycles instead of long continuous holding tasks. The system minimizes unnecessary movement, reduces prolonged arm elevation, and structures activities so that learners can complete tasks with fewer extended holding periods. Simple timing logic and flow control rules are used to manage interaction pacing.
Tools and platforms with justification	Unity 3D and mobile AR devices are used for building and testing the ergonomic interaction flow because they allow realistic simulation of handheld use conditions. Device motion sensors such as the gyroscope and accelerometer may also be used to study movement behavior during interaction, which is consistent with the input structure described in the system overview.
Data collection and ethical considerations	Data is gathered through observation, short usability sessions, movement patterns, session duration, and child or facilitator feedback related to comfort and tiredness. Because the participants are children, the sessions must be short, safe, and carefully monitored. The study stops immediately if

	discomfort is observed. Ethical procedures such as consent, assent, and anonymization is followed.
Validation strategy and performance metrics	The ergonomic design is evaluated using duration of comfortable use, number of repositioning actions, observed fatigue signs, movement frequency, and perceived comfort ratings from facilitator-assisted feedback. The success of this section depends on whether the new interaction flow reduces strain while preserving usability and learning engagement.

Table 3.4: Methodology: Managing Cognitive Load in Multimodal Learning

Aspect	Methodology
Research approach	This subsection uses a learning-centered design and experimental usability evaluation approach. The focus is to design AR presentation and interaction in a way that reduces unnecessary mental effort for Grade 1 learners.
Technical methods and algorithms	The AR interface applies simple multimedia learning principles such as spatial integration of information, reduced visual clutter, controlled timing of prompts, and limited simultaneous presentation of competing elements. Digital signposts and guided interaction cues are used so that children can focus on the learning task without excessive attention switching.
Tools and platforms with justification	Unity 3D and AR Foundation is used because they allow precise control over the placement, timing, and visibility of AR content. This makes them suitable for testing how virtual labels, prompts, and interactive cues influence attention and cognitive load during storybook-based learning.
Data collection and ethical considerations	Data is collected using usability observations, task performance records, engagement patterns, and simplified post-activity feedback from teachers or facilitators. System logs such as task completion time, hesitation, repeated prompts, and interaction flow interruptions are used. Since the target group is children, the evaluation methods remain age-appropriate, non-stressful, and ethically supervised

Validation strategy and performance metrics	This subsection is validated using indicators such as task completion success, time on task, number of repeated prompts, observed distraction, engagement level, and simplified cognitive load indicators gathered through teacher/facilitator-assisted evaluation. Particular attention is given to whether the design reduces split attention and supports clearer learning interaction.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution and Its Components

The proposed solution is a handheld Augmented Reality (AR) storybook system designed to support Cambridge Primary Science Grade 1 learning in Sri Lankan international schools. The system combines a printed storybook with mobile AR visualizations, interactive learning activities, and child-friendly feedback mechanisms to create an engaging and developmentally appropriate learning experience. When a learner points a mobile device at a storybook page, the system detects the page, overlays relevant virtual content, and enables simple guided interaction through touch and visual prompts. The overall solution is intended to support both learning engagement and concept understanding while maintaining usability, comfort, and age-appropriate interaction.

At the system level, the proposed solution consists of four main components contributed by the four group members. These components work together as an integrated learning system. The first component focuses on **content delivery and educational interaction**, which manages the story flow, lesson progression and presentation of science learning material. The second component focuses on **assessment and learner response handling**, which captures learner actions and supports activity-based evaluation. The third component focuses on **visual AR rendering and object presentation**, which controls how digital learning content is displayed on top of the physical storybook. The fourth component, which is the focus of this study, is the **hybrid interaction framework**, which manages child-friendly mobile interaction, occlusion handling and anchoring stability, handheld ergonomic flow, and cognitive load reduction. Together, these components form a complete AR learning environment.

4.2 System Diagram of Components

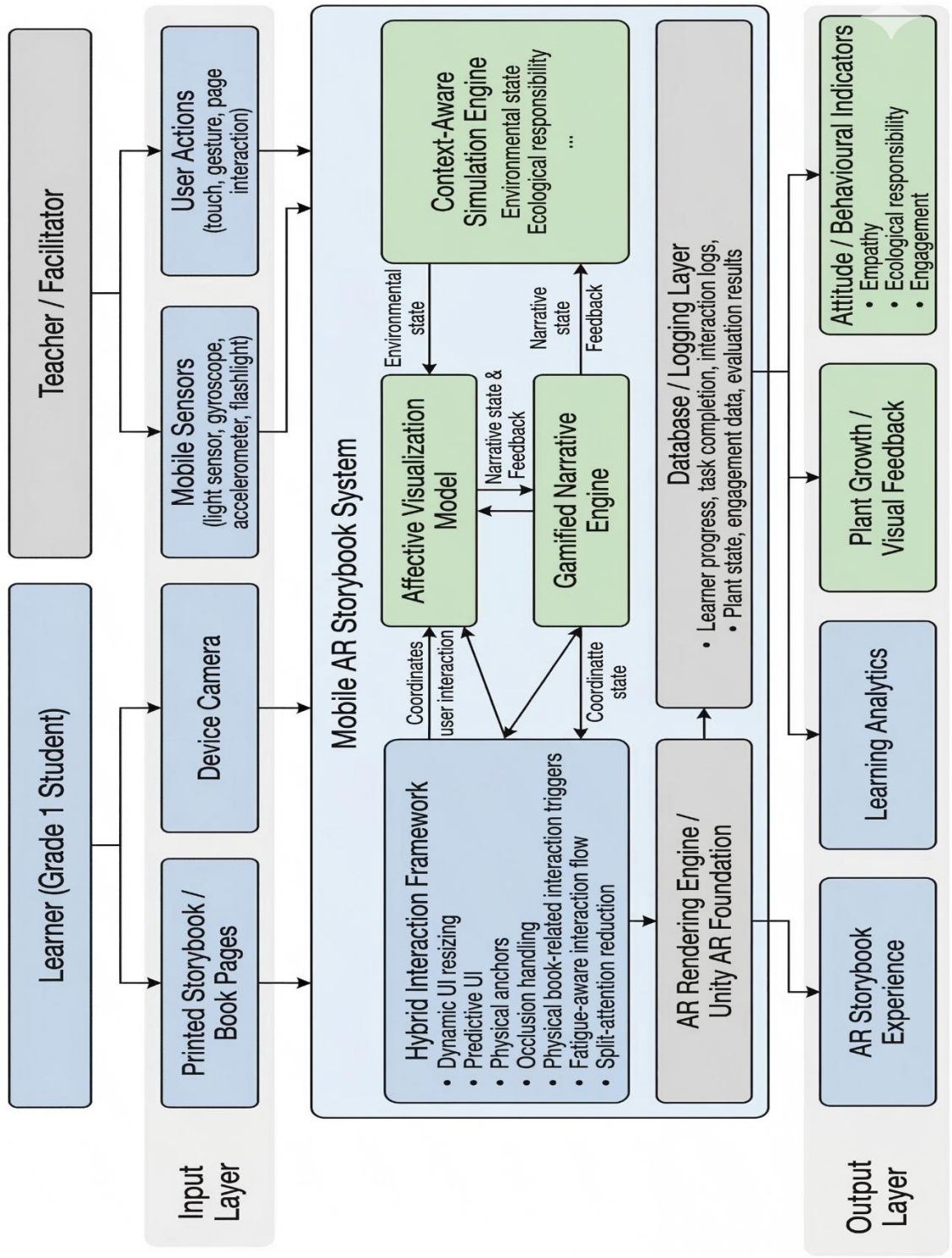


Figure 4.1: *System Diagram for Components*

4.3 Individual Component and Its Parts

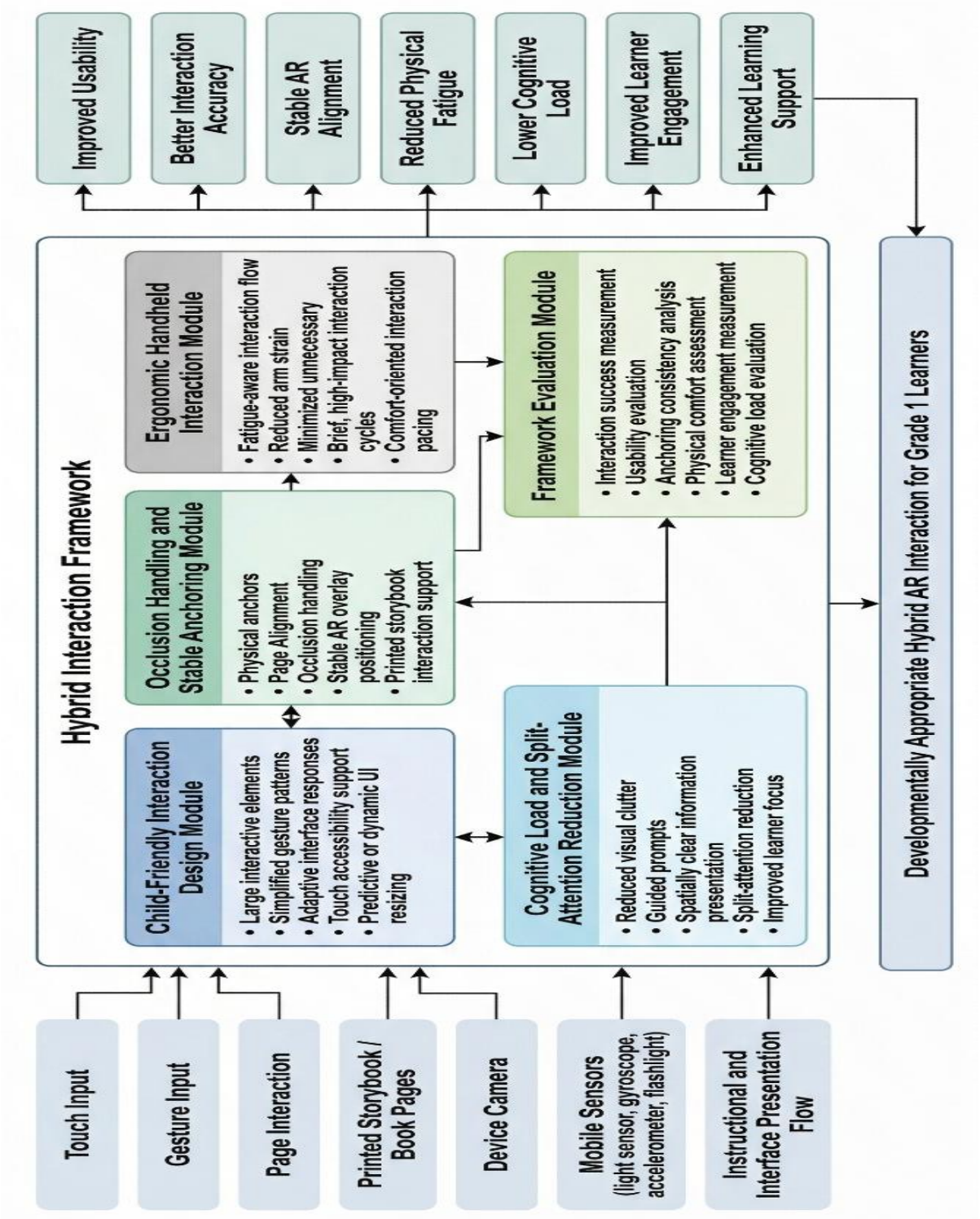


Figure 4.2: *Individual Component and Its Parts*

4.4 Component Interactions

The hybrid interaction framework interacts continuously with the other system components during operation. When the learner opens a storybook page and points the mobile device camera toward it, the AR rendering component detects the page and prepares the related virtual content. At this stage, the hybrid interaction framework manages how the child interacts with the content by controlling touch behavior, interface layout, and prompt delivery. It also supports stable alignment of digital elements with the physical page while allowing natural hand movement and tactile interaction with the book.

The component also communicates with the content delivery and assessment-related modules. For example, when a learner completes an AR task, the interaction framework passes input results, timing information, and behavioral indicators to the relevant evaluation or logging component. At the same time, it receives instructional states and activity flow information from the content module in order to present prompts, feedback, and guided actions at the correct time. In this way, the hybrid interaction framework does not work in isolation; rather, it functions as an intelligent coordination layer that improves the quality of interaction between the learner and the overall AR learning system.

4.5 Data Flow

The data flow of the hybrid interaction framework begins with input capture from the mobile device and the physical storybook environment. The camera captures the printed page, while the touchscreen and device sensors collect user interaction and movement data. These inputs are then processed by the hybrid interaction framework. First, the interaction design module interprets touch input and adjusts interface behavior when needed. Next, the anchoring and occlusion module uses page recognition and AR alignment information to position digital content correctly. Then, the ergonomic module manages the pacing and structure of interaction, while the cognitive load module filters and organizes the presentation of prompts, labels, and interactive elements.

After processing, the system generates outputs in the form of stable AR overlays, child-friendly interface responses, guided actions, and simplified multimodal presentation. At the same time, selected interaction data such as touch attempts, task completion time, anchor stability measures, and usability-related observations can be passed to the

logging or evaluation layer of the overall system. Therefore, the data flow of this hybrid interaction framework moves from camera and sensor input, to interaction and AR processing, and finally to visual output and evaluation data generation. This flow ensures that the learner receives a smooth and manageable AR experience while the system also captures useful evidence for later analysis.

4.6 Scalability and Security Considerations

From a scalability perspective, the hybrid interaction framework should be designed so that it can support additional storybook pages, new learning activities, and future curriculum extensions without requiring major architectural changes. Since the system is developed as a modular handheld AR solution, each internal part of the framework can be improved or expanded independently. For example, new gesture adaptations, improved anchoring methods, or alternative prompt strategies can be introduced without affecting the full system design. This modularity also supports reuse in other early learning contexts beyond the initial science topic.

From a security and privacy perspective, the system must handle learner interaction data carefully, especially because the target users are young children. Any usage logs, observation records, or task-related data collected during testing should be anonymized and stored securely. Personal identifiers should not be attached to the recorded data, and access to collected information should be limited to authorized research purposes only. In addition, the system should avoid collecting unnecessary personal information and should operate in a safe, classroom-appropriate manner. These considerations are important not only for ethical compliance, but also for building a trustworthy educational technology solution for child users.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The primary stakeholder of the proposed system is the **Grade 1 learner**, because the hybrid interaction framework is designed to support the child's direct interaction with the handheld AR storybook. The learner uses the system through touch, gesture, page interaction, and device movement, and therefore the component must be designed around the child's motor ability, comfort, and attention span. The **teacher or facilitator** is another key stakeholder, because classroom guidance, activity supervision, and interpretation of learner performance depend on how clearly and reliably the system supports interaction and feedback. In addition, **parents and school administrators** can be considered supporting stakeholders because they are concerned with the safety, usability, and educational value of the solution for children. These stakeholder roles are consistent with the proposed handheld AR storybook system, which combines printed pages, camera input, sensor data, and child interaction in an educational setting.

5.2 Functional Requirements

The hybrid interaction framework shall support the following functional requirements:

FR1. The system shall detect interaction with printed storybook pages and connect physical page content with the relevant AR content through page-based anchoring. This requirement directly supports the hybrid connection between the physical book and digital overlays described in the solution overview.

FR2. The system shall provide a child-friendly touch interface with large interactive elements and simplified gesture patterns suitable for early learners.

FR3. The system shall dynamically resize or adjust interactive controls in response to the learner's interaction behavior, including instability caused by device shake, in order to improve touch accessibility.

FR4. The system shall support physical anchors and occlusion handling so that the learner can interact naturally with the printed book without important AR overlays being visually blocked by hands or physical contact.

FR5. The system shall maintain stable AR overlay alignment during handheld interaction with the storybook pages. This is necessary because the solution depends on the accurate placement of virtual content over printed pages during user interaction.

FR6. The system shall structure interaction flow into short, manageable activity bursts to reduce prolonged arm use and support comfortable handheld interaction.

FR7. The system shall present digital signposts, prompts, or guided interaction cues that help learners focus on the task and reduce split attention during AR learning.

FR8. The system shall collect interaction data such as touch attempts, task completion, and usability-related behavior for evaluation purposes.

FR9. The system shall support usability evaluation related to cognitive load, engagement, and the effectiveness of the framework in reducing split attention for the target age group.

5.3 Non-Functional Requirements

The non-functional requirements are especially important because they directly affect the success of the Hybrid Interaction Framework.

NFR1. Usability: The system must be highly usable for Grade 1 learners by offering simple, clear, and age-appropriate interaction patterns. For this research, usability affects whether the hybrid interaction framework can successfully reduce mis-taps, support natural page interaction, and help children engage with AR content without confusion. Since the novelty of the framework includes predictive UI, dynamic button resizing, and physical book-related interaction triggers, usability is central to the research contribution.

NFR2. Performance: The system must respond in real time during AR interaction, including page detection, UI adjustment, anchoring, and occlusion handling. For the hybrid interaction framework, poor performance would directly reduce alignment accuracy, make adaptive UI behavior ineffective, and interrupt the flow of interaction. Since the framework depends on immediate feedback and stable AR response, performance has a direct impact on the validity of the study.

NFR3. Reliability and Stability: The system must maintain reliable AR positioning and consistent interaction behavior during repeated use. In the context of this research,

reliability affects whether physical anchors, occlusion handling, and guided interaction remain stable across different learner actions and classroom conditions. If the framework behaves inconsistently, the usability and comfort goals of the research cannot be achieved.

NFR4. Safety and Child Appropriateness: The system must provide a safe and developmentally appropriate interaction experience for children. This directly affects the hybrid interaction framework because the component is specifically designed to reduce physical fatigue, split attention, and interaction difficulty for young learners.

NFR5. Ergonomic Comfort: The system must support comfortable handheld use over the intended learning period. This requirement has a direct relationship with the hybrid interaction framework because one of the core tasks is to prevent the Gorilla Arm effect and reduce physical fatigue through brief, high-impact interaction flow. Therefore, ergonomic quality is not only a general usability concern but a direct success criterion of this research.

NFR6. Cognitive Simplicity: The system must minimize unnecessary cognitive load by reducing visual clutter, split attention, and overly complex interaction sequences. For the hybrid interaction framework, this requirement determines whether it truly supports learner focus and understanding. Since usability testing is intended to measure cognitive load and split attention reduction, this requirement directly shapes how the component is designed and evaluated.

NFR7. Data Privacy and Ethical Compliance: The system must protect participant privacy and handle research data responsibly. This affects the hybrid interaction framework because the component collects interaction logs and usability-related data from children. The research states that parental consent, child assent, anonymization, and institutional ethical compliance are required, which makes privacy protection an essential non-functional requirement of the component evaluation process.

NFR8. Maintainability and Modularity: The system should be modular enough to allow improvement or extension of the interaction framework without redesigning the whole application. This is important for the hybrid interaction framework because its four key areas—interaction design, anchoring and occlusion, ergonomics, and cognitive load management may need iterative refinement during development and testing.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the proposed solution is international schools in Sri Lanka that follow the Cambridge Primary Science curriculum, especially schools that seek interactive and technology-supported learning tools for early primary education. The system is designed as a handheld AR storybook that runs on standard ARCore- and ARKit-enabled smartphones and tablets, which makes it suitable for schools that already use mobile devices or tablets in classroom activities. Although the AR storybook is developed according to the Cambridge Grade 1 Science syllabus in international schools, it is also appropriate for 6-year-old children who enjoy learning about plants, because the concept, visuals, and interaction style are intended for early learners.

Table 6.1: Primary Customer Persona

Category	Details
Persona Name	School Decision-Maker
Role	Principal / Sectional Head / ICT Coordinator / Primary Curriculum Coordinator
Type	Primary Customer
Profile	Responsible for selecting and approving educational technology solutions for the school.
Needs	A reliable, curriculum-aligned, safe, and innovative learning tool that improves student engagement and supports classroom teaching.
Goals	To introduce effective digital learning solutions that enhance educational quality and support modern teaching methods.
Pain Points	Difficulty finding age-appropriate AR tools that are educationally meaningful, easy to implement, and suitable for young learners.
How the System Helps	The AR storybook provides a research-based, child-friendly, and curriculum-relevant learning solution for Grade 1 science education.

Table 6.2: Secondary Customer Persona

Category	Details
Persona Name	Grade 1 Science Teacher
Role	Early Primary Science Teacher
Type	Secondary Customer / Direct User
Profile	Uses the system in classroom teaching to help students understand plant-related science concepts in an interactive way.
Needs	A classroom-friendly tool that is easy to use, engaging for children, and supportive of concept understanding.
Goals	To make science learning more interesting, understandable, and memorable for Grade 1 learners.
Pain Points	Traditional teaching methods may not always keep young learners engaged or help them clearly visualize plant-related concepts.
How the System Helps	The AR storybook combines storytelling, visuals, and guided interaction to make science lessons more engaging and easier to understand.

Table 6.3: Indirect Customer Persona

Category	Details
Persona Name	Parent of a Grade 1 Learner
Role	Parent / Guardian
Type	Indirect Customer
Profile	Supports the child's learning at home and values meaningful educational experiences beyond the classroom.
Needs	A safe, engaging, and educational tool that helps the child learn effectively in a guided or supervised setting.
Goals	To support the child's curiosity, enjoyment, and understanding of science concepts at home.
Pain Points	Limited access to interactive educational tools that are both enjoyable and academically useful for young children.
How the System Helps	The AR storybook supports emotional engagement, guided learning, and progress-related feedback, making it useful for supervised home learning as well.

6.2 Value Proposition

The main value proposition of the proposed system is that it offers a child-centered, safe, and engaging AR learning experience that combines physical storybooks with interactive digital content to improve understanding of plant-related science concepts. The proposed solution already emphasizes that printed storybook pages, camera input, mobile sensor data, and guided interactions are integrated to create an AR learning experience that is more meaningful than static textbook diagrams. In addition, the system is differentiated by its coordinated research modules, particularly the Hybrid Interaction Framework, which addresses touch accessibility, anchoring between physical pages and 3D content, split-attention reduction, occlusion handling, and fatigue-aware interaction support.

From a commercialization perspective, this gives the product clear value for schools: it is not just an AR storybook, but a research-driven educational tool designed specifically for young learners' usability, comfort, and engagement needs. This makes it more attractive than generic AR applications that may focus mainly on visual novelty without considering children's fine motor skills, physical comfort, or cognitive load.

6.3 Revenue Model

A suitable revenue model for this product is a business-to-school licensing model. Under this model, schools would pay an annual or term-based subscription for access to the AR storybook application, learning content, teacher support features, and software updates. This approach is practical because the system is intended for structured educational use rather than one-time casual entertainment. A second possible model is a per-classroom or per-device licensing structure, where schools pay based on the number of student devices or class groups using the system.

An additional revenue stream could come from content expansion packs, such as new storybooks, additional science topics, or curriculum-aligned activity sets for later grades. Since the proposed architecture is modular and includes multiple coordinated research components, the solution has the potential to be extended beyond the initial plant-growth topic into broader environmental or science learning domains. In the longer term, the product could also be offered through partnerships with educational publishers or private tuition and learning centers.

6.4 Cost Estimation

The main cost categories of commercialization would include software development and refinement, content production, 3D asset and animation development, testing and quality assurance, teacher onboarding materials, and maintenance/support costs. The project requires practical development with Unity 3D, AR Foundation, ARCore, and ARKit, as well as ongoing prototyping, iteration, and testing. These factors suggest that the early commercialization stage would involve costs related to continued technical development, optimization, and device compatibility testing.

There would also be costs for producing high-quality storybook pages, visual assets, and classroom-ready deployment materials. If the product is marketed to schools, teacher guides, setup instructions, and user support documentation would also be necessary. In the pilot stage, costs may be relatively high per user because the product is still being refined. However, once the core application and content pipeline are established, the marginal cost of distributing the software to additional schools would be lower than that of physical-only learning tools.

6.5 Pricing Strategy

The pricing strategy should be based on institutional affordability and educational value. A practical approach would be to offer a school subscription package with different tiers depending on the number of users, classes, or content modules required.

Because the target customers are schools rather than individual mass-market consumers, the pricing should remain reasonable enough for adoption while also reflecting the research-based and curriculum-aligned value of the product. A pilot pricing strategy may also be used in the initial stage, where selected schools receive the system at a discounted rate in exchange for implementation feedback and case-study data. This would support both early adoption and product refinement.

6.6 Competitive Advantage

The proposed solution has several competitive advantages. First, it combines physical storytelling with handheld AR in a way that is explicitly designed for young learners rather than adapted from adult-centered AR systems. The Hybrid Interaction Framework includes predictive UI, dynamic button resizing, physical book-related

interaction triggers, and fatigue-related interaction design to improve usability and comfort for children in handheld AR learning. This gives the product a strong advantage over many existing AR learning tools that focus mainly on visual interaction without addressing motor accessibility, ergonomic comfort, or split-attention reduction.

Second, the product is built as an integrated educational ecosystem rather than as a standalone AR animation tool. The complete system includes a Hybrid Interaction Framework, an Affective Visualization Model, a Narrative Engine, and a Context-Aware Simulation Engine working together as coordinated modules. This means the product can offer emotionally responsive visuals, story-driven engagement, and real-world sensor-based interaction in addition to usability improvements. Third, because the solution is intended for early science education and is aligned to a school curriculum context, it has stronger classroom relevance than general-purpose AR entertainment applications.

6.7 Intellectual Property Considerations

The intellectual property (IP) of the proposed product would likely include several protectable elements: the software implementation, the interaction design logic, the storybook content, the 3D assets and animations, and possibly the novel integration of hybrid interaction mechanisms such as adaptive UI resizing, physical anchors, book-related triggers, and fatigue-aware flow design. These aspects may support future claims for software ownership, design protection, or publication-based research recognition.

At the commercialization stage, it will be important to clearly determine the ownership of the software, assets, and content among the student researchers, the university, and any future industry partners. Copyright protection would apply to the storybook design, visuals, animations, and source code, while trademark protection may later be considered for the product name and brand identity. If the final implementation contains sufficiently novel technical methods, there may also be potential for patent-related discussion, although that would require legal and institutional review. In addition, care must be taken to avoid infringement when using third-party libraries, asset packs, or audio resources during product development.

7. BUDGET AND JUSTIFICATION

The proposed project requires resources for hardware procurement, software license, cloud services and human effort to ensure successful design, implementation and evaluation of the hybrid interaction framework. The budget has been carefully estimated to include only essential costs directly contributing to the research outcomes. Table 7.1summarizes the proposed budget, while table 7.2 justifies each item below.

Table 7.1: Estimated Budget for the Research Project

Category	Description	Estimated Cost (LKR)
Hardware Costs	One mid-range smart tablet for AR testing, if a personal device is not sufficient.	35,000
	Printed storybook pages / markers / prototype book materials for repeated testing.	5,000
	Simple device stand / holder / protective cover for testing sessions.	3,000
Software Licenses	Unity Personal, AR Foundation, ARCore, ARKit, Visual Studio Community, Figma basic tools. Graphic/audio asset support using free or student-access tools.	Free
Cloud Services	Google Drive / OneDrive free tier for backup, documentation, research files, logs and test data.	3,000
Human Effort Cost	Contingency allocation for minor development and testing expenses.	8,000
	Total Estimated Cost	**54,000**

Table 7.2: Budget Justification for the Research Project

Category	Justification
Hardware costs	The proposed solution depends on a handheld mobile AR environment using printed storybook pages, camera input, and mobile sensor data such as light, gyroscope, and accelerometer. Therefore, at least one compatible mobile device and printed prototype material are needed for reliable testing.
Software licenses	The technical stack includes Unity 3D, AR Foundation, ARCore, and ARKit, is free under student licenses.
Cloud services	Cloud cost is free because GitHub Student Developer Pack provides cloud storage, ensuring no additional cost for Cloud Services.
Human effort cost	Human effort cost is limited to practical research expenses such as travel, printing, and participant-session preparation rather than paid development staff. This is reasonable because the research is a student-led prototype study, but some cost is still needed for ethical documentation, printed material, school coordination, and pilot evaluation activities involving children.

8. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) of the proposed research is organized according to the four integrated components of the handheld AR storybook system. In the proposed solution, the system operates through printed storybook pages, camera and sensor inputs, and user actions, while the process layer is divided into four coordinated modules: the Hybrid Interaction Framework, the Affective Visualization Model, the Gamified Narrative Engine, and the Context-Aware Simulation Engine. The final outputs are expected to include the AR storybook experience, learning analytics, and attitude-related indicators.

Table 8.1: Task Breakdown per Member

Member	Component	Task Breakdown
IT22641410 – Wijayarathne K.N.	Hybrid Interaction Framework	Design and implement the dynamic UI mechanism for children's fine motor support; develop physical anchors and occlusion handling for printed storybook interaction; design fatigue-aware interaction flow to reduce prolonged arm strain; conduct usability testing related to cognitive load, engagement, and split-attention reduction
IT22179258 – Palliyaguruge S.N.	Affective Visualization Model	Design AR-optimized 3D plant assets; create animations for plant health, stress, and recovery; apply visual and audio effects to improve emotional readability; conduct child-centered AR trials to refine interpretation and empathy-oriented visual performance.
IT22237804 – Amarawardhane A.S.	Gamified Narrative Engine	Develop the branching story structure; create AI/rule-based dialogue and feedback according to plant condition and context; implement time-delta persistence for plant growth and recovery; design redemption-based rewards and evaluate environmental attitudes and responsibility.
IT22626974 – Dissanayaka D.M.D.D.	Context-Aware Simulation Engine	Develop light-sensing functionality; create algorithms to simulate plant growth based on environmental conditions and user actions; optimize sensor aggregation for efficiency; implement flashlight-assisted growth support and gyroscope-based wind/water simulation.

Table 8.2: Timeline Alignment

Time Period	Planned Activities	Related Deadline
March 2026	Finalize proposal, refine requirements, confirm component scope, and prepare initial designs for all four modules.	Proposal Submission / Proposal Presentation
April 2026	Begin early prototype development for interaction design, visualization, narrative logic, and simulation logic.	Proposal milestone continuation
May 2026	Complete first-stage implementation of core module functions and prepare for initial review.	Progress Presentation 1 / Check List Submission
June 2026	Continue implementation and begin cross-component integration, including AR page flow, module communication, and output handling.	Post-Progress 1 development phase
July 2026	Conduct usability testing, debugging, integration refinement, and preliminary evaluation of the overall prototype.	Preparation for Progress Presentation 2
August 2026	Complete the full product, finalize working prototype, and ensure all four modules function together as one integrated system	Progress Presentation 2
September 2026	Improve stability, finalize evaluation, organize documentation, and prepare report and website content.	Final documentation phase
October 2026	Submit final report, website, final checklist, and complete final presentation and viva.	Final submission milestones
December 2026	Prepare and submit research paper publication evidence	Research paper publication evidence submission

Table 8.3: Balanced Workload Distribution

Work Area	Workload Distribution Explanation
Component Ownership	Each member is assigned one major research component, which creates clear responsibility and avoids unnecessary overlap. This supports balanced distribution across interaction, visualization, narrative, and simulation development.
Technical Development	The work is balanced because all four members develop a comparable subsystem with its own design, implementation, and testing responsibilities.
Integration Work	Integration is shared because the final system depends on interaction among all four modules inside one AR storybook platform. This includes combining sensor input, page tracking, story flow, visualization, and usability support.
Testing and Evaluation	Evaluation is also shared, since the project collects observation data, interaction logs, plant growth status, and pre/post outcomes, and requires ethical handling of child participants.
Final Deliverables	Report writing, presentation preparation, website completion, and final refinement are group responsibilities, ensuring that no single member carries all final-stage work.

9. GANTT CHART

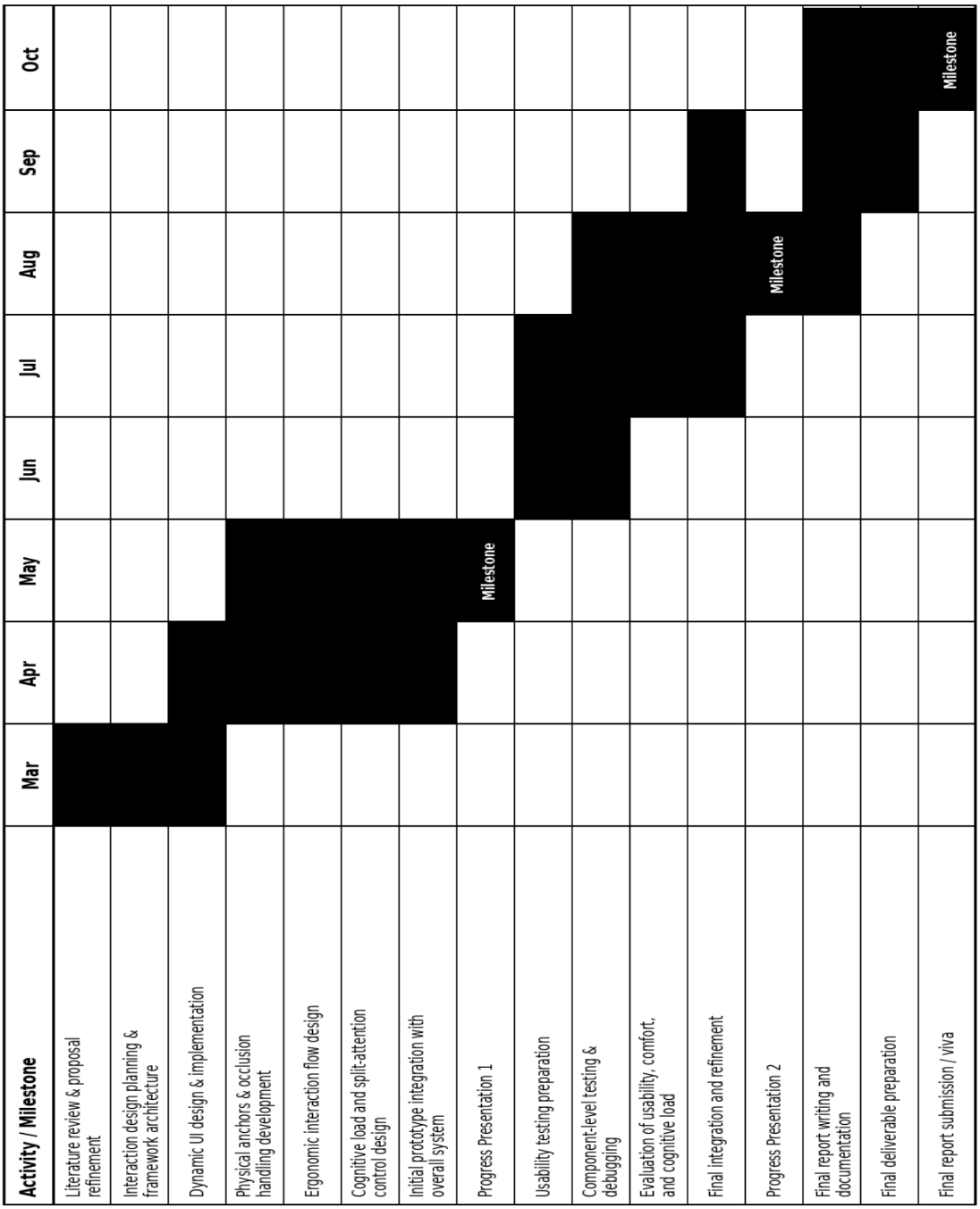


Figure 9.1: *Gantt Chart for the Hybrid Interaction Framework*

This Gantt chart shows that the Hybrid Interaction Framework begins with requirement analysis, design planning, and literature-based refinement during the proposal stage. Development work then focusses on four key technical areas: dynamic UI support,

physical anchoring and occlusion handling, ergonomic interaction flow, and cognitive load reduction. The middle phase of the schedule is allocated for implementation, integration, and testing, while the later phase is dedicated to evaluation, refinement, documentation, and final submission. This timeline ensures that the component is completed in alignment with the full research project schedule and is ready for integration into the complete AR storybook system before the final presentation.

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11. APPENDICES

11.1 Appendix A – Survey Instruments

This appendix presents the sample instruments that will be used to collect data during the evaluation of the proposed handheld AR storybook system. Since the study involves Grade 1 learners, direct questionnaires will be kept simple and supported by teacher or facilitator observation. The main instruments will include a structured observation sheet, a teacher/facilitator feedback form, and a simple post-activity evaluation format.

A.1 Observation Sheet

The observation sheet will be used during child interaction sessions to record behavior related to usability, engagement, and physical comfort.

Table 11.1: Observation sheet

Observation Area	Description
Interaction Accuracy	Whether the child can tap, select, and interact with the AR interface successfully.
Engagement Level	Whether the child remains interested and attentive during the activity
Physical Comfort	Whether the child shows signs of discomfort, fatigue, or difficulty in holding the device
Attention Behavior	Whether the child is confused by split attention or distracted by interface elements
Task Completion	Whether the child completes the given AR learning activity successfully

A.2 Teacher / Facilitator Feedback Form

The teacher or facilitator will be asked to evaluate the usability and educational value of the system.

- The AR storybook was easy for the child to use.
- The interaction design was appropriate for the age group.

- The AR content remained stable and understandable during use.
- The child appeared comfortable during the activity.
- The system supported learning and engagement effectively.

A.3 Post-Activity Evaluation

This will include a short teacher-assisted evaluation of learner response after the session, focusing on:

- level of participation
- understanding of the science concept
- ability to follow the interaction flow
- interest in repeating the activity

11.2 Appendix B – Detailed Diagrams

This appendix contains the detailed technical diagrams used to explain the structure and operation of the proposed system.

The appendix may include:

- Overall System Architecture Diagram
- Hybrid Interaction Framework Architecture Diagram
- Gantt Chart / Work Plan Diagram

B.1 Overall System Architecture

This diagram presents the complete AR storybook system with the four research components:

1. Hybrid Interaction Framework
2. Affective Visualization Model
3. Gamified Narrative Engine
4. Context-Aware Simulation Engine

B.2 Individual Component Diagram

This diagram shows the internal structure of the Hybrid Interaction Framework, including:

- Child-Friendly Interaction Design Module
- Occlusion Handling and Stable Anchoring Module
- Ergonomic Handheld Interaction Module
- Cognitive Load and Split-Attention Reduction Module
- Framework Evaluation Module

11.3 Appendix C – Preliminary UI Mockups

This appendix includes the early visual designs of the proposed handheld AR storybook system. These mockups are intended to demonstrate the planned interaction style and user interface structure before full implementation.

C.1 Planned UI Design Features

The preliminary UI is expected to include:

- large interactive buttons
- minimal text
- simple prompts
- child-friendly color use
- reduced visual clutter
- guided interaction indicators

C.2 Purpose of Mockups

These mockups are included to show how the interface design supports:

- touch accessibility for young learners
- easier navigation
- reduced cognitive load
- better learner focus during AR interaction

11.4 Appendix D – Risk Analysis

This appendix presents the main risks associated with the research and the proposed mitigation strategies.

Table 11.2: Risk Analysis

Risk	Description	Mitigation Strategy
Unstable AR Tracking	AR overlays may shift or lose alignment during use	Use page-based anchoring, controlled testing conditions, and repeated calibration
Occlusion Errors	Hands or book interaction may block virtual content incorrectly	Improve occlusion handling logic and test with natural interaction patterns
Interaction Difficulty	Children may struggle with touch input or gestures	Use large buttons, simplified gestures, and adaptive interface design
Physical Fatigue	Children may experience discomfort during prolonged handheld use	Design short interaction cycles and reduce unnecessary movement
Cognitive Overload	Too many visual elements may distract the learner	Reduce clutter, use guided prompts, and apply split-attention reduction principles
Limited Participant Access	Difficulty in arranging child-centered testing sessions	Coordinate early with schools and prepare alternative pilot sessions
Data Collection Delays	Delays in observations, forms, or test data gathering	Prepare instruments early and maintain a structured evaluation schedule

11.5 Appendix E – Ethical Documents

This appendix includes the ethical and participant-related documents required for the research study, especially because the target participants are young children.

The appendix may include:

- Parent / Guardian Consent Form
- Child Assent Form
- Participant Information Sheet
- Privacy and Confidentiality Statement
- Ethical Clearance Submission Documents

E.1 Parent / Guardian Consent Form

This document will explain:

- the purpose of the study
- the child's involvement
- the type of activities to be carried out
- the voluntary nature of participation
- privacy protection measures

E.2 Child Assent Form

This will provide a simple explanation, in age-appropriate language, to confirm that the child is willing to participate.

E.3 Privacy and Confidentiality

All collected data will be anonymized. No unnecessary personal information will be stored, and all research data will be used only for academic purposes.